

Nice 📖 revision time — good move before moving ahead.

I'll give you a **clean, exam-oriented REVISION of UNIT-1, short + clear**, exactly matching what you studied.

You can read this **one time before exams**.

🔗 UNIT-1 REVISION (Python & IoT – Raspberry Pi)

1. Internet of Things (IoT)

- IoT = Network of physical devices connected to the internet
- Devices collect and exchange data automatically
- Uses sensors, internet, and software

Examples: Smart home, smart city, smart agriculture

2. Applications of IoT

- Smart Home
- Healthcare
- Agriculture
- Smart Cities
- Industrial Automation
- Transportation

🔗 Used for automation, monitoring, and control

3. Advantages of IoT

- Automation
- Time saving
- Remote monitoring
- Real-time data
- Improved efficiency

Disadvantages of IoT

- Security issues
 - Privacy problems
 - High initial cost
 - Internet dependency
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4. Raspberry Pi-3 Components

- CPU (Processor)
 - RAM (1 GB)
 - USB ports
 - HDMI port
 - Ethernet
 - Wi-Fi & Bluetooth
 - GPIO pins (40)
 - SD card slot
 - Power supply
 - Camera & Display interface
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5. GPIO Pins

- GPIO = General Purpose Input Output
 - Used to connect sensors and actuators
 - Works at **3.3V**
 - Types:
 - Power pins (5V, 3.3V, GND)
 - Input/Output pins
 - Special pins (UART, I2C, SPI)
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6. Build PC Using Raspberry Pi

Required:

- Raspberry Pi
- SD card with OS
- Monitor (HDMI)
- Keyboard & Mouse
- Power supply

Used for programming, browsing, IoT projects

7. Install OS to Raspberry Pi

- Raspberry Pi OS written to SD card
 - Uses Raspberry Pi Imager
 - SD card acts as hard disk
 - OS required to run Python programs
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8. Familiarize to Python Programming Language

- Python is high-level, interpreted language
 - Simple syntax
 - Uses indentation
 - Case sensitive
 - Easy for beginners
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9. Different Versions of Python

- Python 2 → Old, discontinued
- Python 3 → Latest, supported

👉 Raspberry Pi uses **Python 3**

10. Python vs Java

Python	Java
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Simple syntax	Complex syntax
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Interpreted	Compiled + Interpreted
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Easy to learn	Harder
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Used in IoT	Used in enterprise apps
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11. Features of Python

- Simple & easy
 - Interpreted
 - High-level
 - Open source
 - Portable
 - Object-oriented
 - Large library
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12. Setting Execution Environment

Requires:

- OS
- Python interpreter
- IDE (IDLE / Thonny)

Python comes pre-installed in Raspberry Pi.

13. Ways to Execute Python Script

1. Interactive mode
2. Script mode
3. Using IDLE
4. Using Command Prompt

5. Using IDE (Thonny)

✓ Exam Tip

- Learn **definitions**
- Remember **tables (Python vs Java)**
- Write **short & neat answers**
- Use **headings**